TASK 7: SQL QUERY IMPLEMENTATION FOR BAKERY DATABASE

STUDENT DETAILS

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SCENARIO

The Artisan Crumb Bakery & Pastry Management System has collected customer, bakery item, order, inventory, payment, and review data. The management team wants to retrieve useful information from the database for daily operations and business analysis. Students will apply SQL queries to search, filter, sort, and generate reports from the existing Bakery Database.

1. Perform Basic SQL Queries

Concept

SQL **SELECT** command is used to retrieve data from database tables.

1. Display all customer details.

```
SQL> SELECT *
  2  FROM Customer;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                          PHONE CITY                CREATED_A
------------------------------ ---------- -------------------- ---------
C001                 John                 Doe
john.doe@email.com                                           27-OCT-26

C002                 Jane                 Smith
jane.smith@email.com                                         27-OCT-26

C003                 Alice                Brown
alice.brown@email.com                                        27-OCT-26


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                          PHONE CITY                CREATED_A
------------------------------ ---------- -------------------- ---------
C004                 Bob                  White
bob.white@email.com                                          27-OCT-26

C005                 Charlie              Green
charlie.green@email.com                                      27-OCT-26
```

2. Display all available bakery items.

```
SQL> SELECT *
  2  FROM Product;

PRODUCT_ID TITLE                    CATEGORY_ID     PRICE SHELF_LIFE_DAYS
---------- ------------------------ ----------- -------- ---------------
P006       Special Wedding Cake     CAT-02        1200.00               3
P001       Chocolate Fudge Cake     CAT-02         750.00               2
P002       Sourdough Bread          CAT-03         350.00               4
P003       Red Velvet Pastry        CAT-01         450.00               2
P004       Croissants                              200.00
P005       Black Forest Cake        CAT-02         600.00               3

6 rows selected.
```

3. Retrieve bakery item names and prices only.

```
SQL> SELECT Title, Price
  2  FROM Product;

TITLE
------------------------------
     PRICE
----------
Special Wedding Cake
      1200

Chocolate Fudge Cake
       750

Sourdough Bread
       350


TITLE
------------------------------
     PRICE
----------
Red Velvet Pastry
       450

Croissants
       200

Black Forest Cake
       600


6 rows selected.
```

4. Display all bakery orders placed by customers.

```
SQL> SELECT *
  2  FROM Orders;

ORDER_ID             CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT
-------------------- -------------------- --------- ------------
ORDER_STATUS
--------------------------------------------------
ORD-505              CUST-104             24-MAR-26         1250
Delivered

ORD-506              CUST-102             25-MAR-26          720
Delivered

ORD-507              CUST-103             26-MAR-26         1450
Baking


ORDER_ID             CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT
-------------------- -------------------- --------- ------------
ORDER_STATUS
--------------------------------------------------
ORD-508              CUST-101             27-MAR-26          890
Pending
```

5. Retrieve payment details.

```
SQL> SELECT *
  2  FROM Payment;

PAYMENT_ID           ORDER_ID             PAYMENT_D
-------------------- -------------------- ---------
PAYMENT_MODE
--------------------------------------------------
PAYMENT_STATUS                           TRANSACTION_AMOUNT
---------------------------------------- ------------------
PAY-107              ORD-507              26-MAR-26
Debit Card
Failed                                              1450

PAY-108              ORD-507              26-MAR-26
Credit Card
Completed                                           1450

PAYMENT_ID           ORDER_ID             PAYMENT_D
-------------------- -------------------- ---------
PAYMENT_MODE
--------------------------------------------------
PAYMENT_STATUS                           TRANSACTION_AMOUNT
---------------------------------------- ------------------

PAY-109              ORD-508              27-MAR-26
Debit Card
Completed                                            890

PAY-101              ORD-501              18-MAR-26
Credit Card

PAYMENT_ID           ORDER_ID             PAYMENT_D
-------------------- -------------------- ---------
PAYMENT_MODE
--------------------------------------------------
PAYMENT_STATUS                           TRANSACTION_AMOUNT
---------------------------------------- ------------------
Completed                                            660

PAY-102              ORD-502              19-MAR-26
UPI
Completed                                            850

PAY-103              ORD-503              20-MAR-26
```

```
PAYMENT_ID          ORDER_ID            PAYMENT_D
------------------- ------------------- ---------
PAYMENT_MODE
-------------------------------------------------
PAYMENT_STATUS                                TRANSACTION_AMOUNT
--------------------------------------------- ------------------
Debit Card
Completed                                                    360

PAY-104             ORD-504             22-MAR-26
Net Banking
Completed                                                    510


PAYMENT_ID          ORDER_ID            PAYMENT_D
------------------- ------------------- ---------
PAYMENT_MODE
-------------------------------------------------
PAYMENT_STATUS                                TRANSACTION_AMOUNT
--------------------------------------------- ------------------
PAY-105             ORD-505             24-MAR-26
UPI
Completed                                                   1250

PAY-106             ORD-506             25-MAR-26
COD
Completed                                                    720

PAYMENT_ID          ORDER_ID            PAYMENT_D
------------------- ------------------- ---------
PAYMENT_MODE
-------------------------------------------------
PAYMENT_STATUS                                TRANSACTION_AMOUNT
--------------------------------------------- ------------------


9 rows selected.
```

2. Apply Filtering Conditions Using WHERE

1. Find bakery items with price greater than ₹500.

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price > 500;

PRODUCT_ID TITLE                     CATEGORY_ID     PRICE SHELF_LIFE_DAYS
---------- ------------------------- ------------ -------- ---------------
P006       Special Wedding Cake      CAT-02        1200.00               3
P001       Chocolate Fudge Cake      CAT-02         750.00               2
P005       Black Forest Cake         CAT-02         600.00               3
```

2. Display bakery items available in stock.

```
SQL> SELECT *
  2  FROM Inventory
  3  WHERE Stock_Quantity > 0;

INVENTO PRODUCT SUPPLIE STOCK_QUANTITY STOCK_STATUS    LAST_UPDA
------- ------- ------- -------------- --------------- ---------
INV001  P001                       25 In Stock
INV002  P002                       15 In Stock
INV003  P003                        5 Low Stock
INV004  P004                       30 In Stock
INV006  P006                       10 In Stock
```

3. Display completed orders.

```
SQL> SELECT * FROM Orders WHERE UPPER(Order_Status) = 'COMPLETED';

ORDER_ID             CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT
-------------------- -------------------- --------- ------------
ORDER_STATUS
--------------------------------------------------
ORD-505              CUST-104             24-MAR-26         1250
Completed

ORD-506              CUST-102             25-MAR-26          720
Completed
```

4. Find bakery items with rating above 4.

```
SQL> SELECT *
  2  FROM Review
  3  WHERE Rating > 4;

REVIEW_ CUSTOME PRODUCT     RATING
------- ------- ------- ----------
REVIEW_TEXT
----------------------------------------
REVIEW_DA
---------
REV001  C001    P001             5
Excellent Chocolate Fudge Cake
20-JAN-26

REV003  C003    P003             5
Delicious Red Velvet Pastry
01-FEB-26

REVIEW_ CUSTOME PRODUCT     RATING
------- ------- ------- ----------
REVIEW_TEXT
----------------------------------------
REVIEW_DA
---------

REV005  C005    P005             5
Excellent Black Forest Cake
10-FEB-26
```

3. Sort Data Using ORDER BY

Tasks

1. Display bakery items from lowest to highest price.

```
SQL> SELECT *
  2  FROM Product
  3  ORDER BY Price ASC;

PRODUCT_ID TITLE                    CATEGORY_ID     PRICE SHELF_LIFE_DAYS
---------- ------------------------ ----------- -------- ---------------
P004       Croissants                            200.00
P002       Sourdough Bread          CAT-03        350.00               4
P003       Red Velvet Pastry        CAT-01        450.00               2
P005       Black Forest Cake        CAT-02        600.00               3
P001       Chocolate Fudge Cake     CAT-02        750.00               2
P006       Special Wedding Cake     CAT-02       1200.00               3

6 rows selected.
```

2. Display customers alphabetically.

```
SQL> SELECT *
  2  FROM Customer
  3  ORDER BY First_Name, Last_Name;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                          PHONE CITY                 CREATED_A
------------------------------ ---------- -------------------- ---------
C003                 Alice                Brown
alice.brown@email.com                                        27-OCT-26

C004                 Bob                  White
bob.white@email.com                                          27-OCT-26

C005                 Charlie              Green
charlie.green@email.com                                      27-OCT-26


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                          PHONE CITY                 CREATED_A
------------------------------ ---------- -------------------- ---------
C002                 Jane                 Smith
jane.smith@email.com                                         27-OCT-26

C001                 John                 Doe
john.doe@email.com                                           27-OCT-26
```

3.Find top expensive bakery items.

```
SQL> SELECT *
  2  FROM (
  3      SELECT *
  4      FROM Product
  5      ORDER BY Price DESC
  6  )
  7  WHERE ROWNUM <= 5;

PRODUCT_ID TITLE                     CATEGORY_ID     PRICE SHELF_LIFE_DAYS
---------- ------------------------ ----------- -------- ---------------
P006       Special Wedding Cake     CAT-02        1200.00               3
P001       Chocolate Fudge Cake     CAT-02         750.00               2
P005       Black Forest Cake        CAT-02         600.00               3
P003       Red Velvet Pastry        CAT-01         450.00               2
P002       Sourdough Bread          CAT-03         350.00               4
```

4. Display latest orders first.

```
SQL> SELECT *
  2  FROM Orders
  3  ORDER BY Order_Date DESC;

ORDER_ID             CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT
-------------------- -------------------- --------- ------------
ORDER_STATUS
--------------------------------------------------
ORD-508              CUST-101             27-MAR-26          890
Pending

ORD-507              CUST-103             26-MAR-26         1450
Baking

ORD-506              CUST-102             25-MAR-26          720
Completed


ORDER_ID             CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT
-------------------- -------------------- --------- ------------
ORDER_STATUS
--------------------------------------------------
ORD-505              CUST-104             24-MAR-26         1250
Completed
```

4. Retrieve Unique Values Using DISTINCT

Tasks

1. Display unique bakery categories.

```
SQL> SELECT DISTINCT Category_Name
  2  FROM Category;

CATEGORY_NAME
--------------------------------------------------
Artisan Breads
Celebration Cakes
Morning Pastries
```

2. Find different payment methods used by customers.

```
SQL> SELECT DISTINCT Payment_Mode
  2  FROM Payment;

PAYMENT_MODE
----------------------------------------
Credit Card
Net Banking
COD
Debit Card
UPI
```

5. Search Products Based on Conditions

Tasks

1. Find bakery items within a specific price range.

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price BETWEEN 200 AND 1000;

PRODUCT_ID TITLE                    CATEGORY_ID     PRICE SHELF_LIFE_DAYS
---------- ------------------------ ----------- -------- ---------------
P001       Chocolate Fudge Cake     CAT-02         750.00               2
P002       Sourdough Bread          CAT-03         350.00               4
P003       Red Velvet Pastry        CAT-01         450.00               2
P004       Croissants                              200.00
P005       Black Forest Cake        CAT-02         600.00               3
```

2. Display bakery items belonging to a particular category.

```
SQL> SELECT P.*
  2  FROM Product P
  3  JOIN Category C ON P.Category_ID = C.Category_ID
  4  WHERE C.Category_Name = 'Celebration Cakes';

PRODUCT_ID TITLE                    CATEGORY_ID     PRICE SHELF_LIFE_DAYS
---------- ------------------------ ----------- -------- ---------------
P006       Special Wedding Cake     CAT-02       1200.00               3
P001       Chocolate Fudge Cake     CAT-02        750.00               2
P005       Black Forest Cake        CAT-02        600.00               3
```

3. Find bakery items currently available in inventory.

```
SQL> SELECT P.Product_ID,
  2         P.Title,
  3         I.Stock_Quantity
  4  FROM Product P
  5  JOIN Inventory I ON P.Product_ID = I.Product_ID
  6  WHERE I.Stock_Quantity > 0;

PRODUCT_ID TITLE                    STOCK_QUANTITY
---------- ------------------------ --------------
P006       Special Wedding Cake                 10
P001       Chocolate Fudge Cake                 25
P002       Sourdough Bread                      15
P003       Red Velvet Pastry                     5
P004       Croissants                           30
```

4. Search bakery items using bakery item name.

```
SQL> SELECT *
  2  FROM Product
  3  WHERE UPPER(Title) LIKE '%CAKE%';

PRODUCT_ID TITLE                    CATEGORY_ID     PRICE SHELF_LIFE_DAYS
---------- ------------------------ ----------- -------- ---------------
P006       Special Wedding Cake     CAT-02       1200.00               3
P001       Chocolate Fudge Cake     CAT-02        750.00               2
P005       Black Forest Cake        CAT-02        600.00               3
```

5. Find low-stock bakery items.

```
SQL> SELECT *
  2  FROM Inventory
  3  WHERE Stock_Quantity < 10;

INVENTO PRODUCT SUPPLIE STOCK_QUANTITY STOCK_STATUS    LAST_UPDA
------- ------- ------- -------------- --------------- ---------
INV005  P005                         0 Out of Stock
INV003  P003                         5 Low Stock
```

6. Retrieve Customer and Product Information

Tasks

1. Display customer details with their orders.

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         C.Last_Name,
  4         O.Order_ID,
  5         O.Order_Date
  6  FROM Customer C
  7  JOIN Orders O ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
ORDER_ID             ORDER_DAT
-------------------- ---------
C004                 Bob                  White
ORD-505              24-MAR-26

C002                 Jane                 Smith
ORD-506              25-MAR-26

C003                 Alice                Brown
ORD-507              26-MAR-26


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
ORDER_ID             ORDER_DAT
-------------------- ---------
C001                 John                 Doe
ORD-508              27-MAR-26
```

2. Display bakery item details with category information.

```
SQL> SELECT P.Product_ID,
  2         P.Title,
  3         P.Price,
  4         C.Category_Name
  5  FROM Product P
  6  JOIN Category C ON P.Category_ID = C.Category_ID;

PRODUCT
-------
TITLE
--------------------------------------------------------------------
     PRICE
----------
CATEGORY_NAME
--------------------------------------------------------------------
P006
Special Wedding Cake
      1200
Celebration Cakes


PRODUCT
-------
TITLE
--------------------------------------------------------------------
     PRICE
----------
CATEGORY_NAME
--------------------------------------------------------------------
P001
Chocolate Fudge Cake
       750
Celebration Cakes


PRODUCT
-------
TITLE
--------------------------------------------------------------------
     PRICE
----------
CATEGORY_NAME
--------------------------------------------------------------------
P005
Black Forest Cake
       600
Celebration Cakes
```

3. Find customers who purchased a specific bakery item.

```
SQL> SELECT DISTINCT
  2         C.Customer_ID,
  3         C.First_Name,
  4         C.Last_Name
  5  FROM Customer C
  6  JOIN Orders O ON C.Customer_ID = O.Customer_ID
  7  JOIN Order_Details OD ON O.Order_ID = OD.Order_ID
  8  JOIN Product P ON OD.Product_ID = P.Product_ID
  9  WHERE P.Title = 'Chocolate Fudge Cake';

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
C002                 Jane                 Smith
```

4. Display bakery items purchased by each customer.

```
SQL> SELECT C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  2         P.Title
  3  FROM Customer C
  4  JOIN Orders O ON C.Customer_ID = O.Customer_ID
  5  JOIN Order_Details OD ON O.Order_ID = OD.Order_ID
  6  JOIN Product P ON OD.Product_ID = P.Product_ID
  7  ORDER BY Customer_Name;

CUSTOMER_NAME
--------------------------------------
TITLE
------------------------------------------------------------------------
Bob White
Black Forest Cake

Jane Smith
Chocolate Fudge Cake
```

7. Apply Multiple Filtering Conditions

Tasks

1. Find cake products costing more than ₹1000

```
SQL> SELECT P.*
  2  FROM Product P
  3  JOIN Category C ON P.Category_ID = C.Category_ID
  4  WHERE C.Category_Name = 'Celebration Cakes'
  5    AND P.Price > 1000;

PRODUCT_ID TITLE                     CATEGORY_ID     PRICE SHELF_LIFE_DAYS
---------- ------------------------- ----------- -------- ---------------
P006       Special Wedding Cake      CAT-02        1200.00               3
```

2. Search bakery items containing the word "Cake".

```
SQL> SELECT *
  2  FROM Product
  3  WHERE UPPER(Title) LIKE '%CAKE%';

PRODUCT_ID TITLE                    CATEGORY_ID     PRICE SHELF_LIFE_DAYS
---------- ------------------------ ----------- -------- ---------------
P006       Special Wedding Cake     CAT-02        1200.00               3
P001       Chocolate Fudge Cake     CAT-02         750.00               2
P005       Black Forest Cake        CAT-02         600.00               3
```

3. Find orders placed within a specific date range.

```
SQL> SELECT *
  2  FROM Orders
  3  WHERE Order_Date BETWEEN TO_DATE('2026-01-01', 'YYYY-MM-DD')
  4                       AND TO_DATE('2026-12-31', 'YYYY-MM-DD');

ORDER_ID             CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-------------------- -------------------- --------- ------------ ----------------------
ORD-505              C004                 24-MAR-26         1250 Completed
ORD-506              C002                 25-MAR-26          720 Completed
ORD-507              C003                 26-MAR-26         1450 Baking
ORD-508              C001                 27-MAR-26          890 Pending
```

8. Generate Basic Business Reports

Report 1: Product Availability Report

Display:

- Product Name
- Price
- Stock Quantity
- Availability Status

```
SQL> SELECT P.Title,
  2         P.Price,
  3         I.Stock_Quantity,
  4         I.Stock_Status
  5  FROM Product P
  6  JOIN Inventory I ON P.Product_ID = I.Product_ID;

TITLE                       PRICE STOCK_QUANTITY STOCK_STATUS
------------------------ -------- -------------- ---------------
Black Forest Cake          600.00              0 Out of Stock
Chocolate Fudge Cake       750.00             25 In Stock
Sourdough Bread            350.00             15 In Stock
Red Velvet Pastry          450.00              5 Low Stock
Croissants                 200.00             30 In Stock
Special Wedding Cake      1200.00             10 In Stock

6 rows selected.
```

Report 2: Customer Report

Total Number of Customers

```
SQL> SELECT COUNT(*) AS Total_Customers
  2  FROM Customer;

TOTAL_CUSTOMERS
---------------
              5
```

New Customer Registrations

```
SQL> SELECT COUNT(*) AS New_Customers
  2  FROM Customer
  3  WHERE Created_At >= ADD_MONTHS(SYSDATE, -1);

NEW_CUSTOMERS
-------------
            5
```

Report 3: Order Report

Total Orders

```
SQL> SELECT COUNT(*) AS Total_Orders
  2  FROM Orders;

TOTAL_ORDERS
------------
           4
```

Completed Orders

```
SQL> SELECT COUNT(*) AS Completed_Orders
  2  FROM Orders
  3  WHERE Order_Status = 'Completed';

COMPLETED_ORDERS
----------------
               2
```

Pending Orders

```
SQL> SELECT COUNT(*) AS Pending_Orders
  2  FROM Orders
  3  WHERE Order_Status = 'Pending';

PENDING_ORDERS
--------------
             1
```

Cancelled Orders

```
SQL> SELECT COUNT(*) AS Cancelled_Orders
  2  FROM Orders
  3  WHERE Order_Status = 'Cancelled';

CANCELLED_ORDERS
----------------
               0
```

Report 4: Product Performance Report

Highest-Priced Bakery Items

```
SQL> SELECT *
  2  FROM (
  3      SELECT *
  4      FROM Product
  5      ORDER BY Price DESC
  6  )
  7  WHERE ROWNUM <= 5;

PRODUCT_ID TITLE                     CATEGORY_ID     PRICE SHELF_LIFE_DAYS
---------- ------------------------- ----------- -------- ---------------
P006       Special Wedding Cake      CAT-02       1200.00               3
P001       Chocolate Fudge Cake      CAT-02        750.00               2
P005       Black Forest Cake         CAT-02        600.00               3
P003       Red Velvet Pastry         CAT-01        450.00               2
P002       Sourdough Bread           CAT-03        350.00               4
```

Most Reviewed Bakery Items

```
SQL> SELECT P.Title,
  2         COUNT(R.Review_ID) AS Review_Count
  3  FROM Product P
  4  JOIN Review R ON P.Product_ID = R.Product_ID
  5  GROUP BY P.Product_ID, P.Title
  6  ORDER BY Review_Count DESC;

TITLE                     REVIEW_COUNT
------------------------- ------------
Chocolate Fudge Cake                 1
Sourdough Bread                      1
Croissants                           1
Black Forest Cake                    1
Red Velvet Pastry                    1
```

Available Bakery Items

```
SQL> SELECT P.Title,
  2         I.Stock_Quantity,
  3         I.Stock_Status
  4  FROM Product P
  5  JOIN Inventory I ON P.Product_ID = I.Product_ID
  6  WHERE I.Stock_Quantity > 0;

TITLE                     STOCK_QUANTITY STOCK_STATUS
------------------------- -------------- ---------------
Special Wedding Cake                  10 In Stock
Chocolate Fudge Cake                  25 In Stock
Sourdough Bread                       15 In Stock
Red Velvet Pastry                      5 Low Stock
Croissants                            30 In Stock
```